Lucas S. Krippendorff

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EDUCATION

University of Pennsylvania — Philadelphia, PA Aug 2024 – May 2028
B.S. Electrical Engineering

EXPERIENCE

SubVysion (YC S26) — Philadelphia, PA · *Hardware Engineer Intern* Mar 2026 – Present
Early-stage startup · Part-time during academic year, full-time Summer 2026

- First engineering hire and hardware lead; responsible for the physical and electrical layer of the product
- Designed and built a four-wheeled autonomous outdoor rover in under two weeks to replace a manual sensor cart, housing a full sensor suite
- Designed the electrical system: power distribution from supply to motors, onboard computer, and sensors, including buck converters, power budgeting, and cable management
- Led mechanical CAD (wheels, sensor mounts, electrical panel, structural components) and fabricated parts via 3D printing and rapid prototyping
- Integrated and powered the sensor suite, ensuring electrical compatibility and reliable sensor-to-motor communication

3D Chimera — Miami, FL · *Engineering Intern* Jun – Jul 2025
3D-printing solutions & Bambu Lab reseller

- Engineered a hybrid storage system combining off-the-shelf materials with custom 3D-printed components, cutting cost and assembly complexity
- Repaired and calibrated Bambu Lab X1E and H2D printers for clients, restoring functionality after shipping damage and operational wear
- Prototyped and tested parts with tuned infill and material profiles, evaluating tensile strength, shear strength, and sealing performance

PROJECTS

22nm CMOS Datapath & Memory — UPenn ESE 3700 Spring 2026

- Designed an 8-bit ripple-carry adder and 16×4 SRAM array at the transistor level in Electric VLSI on a 22nm HP PTM process (V_{DD} = 0.8 V)
- Compared NAND-based and transmission-gate XOR2 topologies via Elmore delay modeling and SPICE validation, selecting the optimal sum/carry datapath

Six-Degree-of-Freedom Robotic Arm — Personal Project May – Aug 2025

- Designed and fabricated custom 26:1 cycloidal drives and 5:1 planetary gearboxes with integrated ball bearings, 3D-printed and driven by NEMA-17 steppers
- Built an inverse-kinematics solver and trajectory planner in Python (Robotics Toolbox), streaming step pulses to motors through Arduino firmware in C++
- Developed a JavaScript/Meshcat web interface to set waypoints, save paths, and command the arm in real time

Cat–Dog CNN Classifier — UPenn ENGR 1050 Dec 2024

- Trained a PyTorch CNN on 24,000 labeled images with CUDA GPU acceleration, reaching 97.8% test accuracy; project went beyond the single-layer perceptron the course had covered
- Built the full preprocessing pipeline (resize, normalize, augment) and evaluated model generalization across held-out datasets

COURSEWORK

Spring 2026: ESE 3500 Embedded Systems Lab · ESE 3700 Digital Circuit Design
Fall 2026 (two graduate-level): ESE 5700 Digital IC & VLSI Fundamentals · ESE 5070 Networks & Protocols · CIS 2400 Introduction to Computer Systems

ACTIVITIES

UPenn Aerial Robotics Team (2024–25) · UPenn Mars Rover Club (2025–Present)

SKILLS

Programming: Python (PyTorch, CUDA), C/C++ (embedded firmware), MATLAB, JavaScript/HTML/CSS, Git
Hardware: Microcontrollers (Arduino, Raspberry Pi), stepper/servo control, power distribution & buck converters, power budgeting, PCB design (KiCad), SPICE, 3D printing & prototyping
CAD/Tools: SolidWorks, Onshape, Fusion 360, KiCad, Electric VLSI, Meshcat, Ableton Live
Languages: English (native), Japanese (beginner)